A solution of a polynomial equation written only using its coefficients, the four basic arithmetic operators (addition, multiplication, subtraction and division), and rational exponentiation (power to a rational number, such as square, cube, square roots, cube roots and so on) is called a **radical solution**. Abel proved that it is impossible to write a radical solution for general polynomial equation of degree five or more.



We state a few results about polynomial equations that are useful in solving higher degree polynomial equations.

- Every polynomial in one variable is a continuous function from $\mathbb{R}$ to $\mathbb{R}$.
- For a polynomial equation $P(x)=0$ of even degree, $P(x)\to\infty$ as $P(x)\to\pm\infty$. Thus the graph of an even degree polynomial start from left top and ends at right top.
- All results discussed on "graphing functions" in Volume I of eleventh standard textbook can be applied to the graphs of polynomials. For instance, a change in the constant term of a polynomial moves its graph up or down only.
- Every polynomial is differentiable any number of times.
- The real roots of a polynomial equation $P(x)=0$ are the points on the x-axis where the graph of $P(x)=0$ cuts the x-axis.
- If a and b are two real numbers such that $P(a)$ and $P(b)$ are of opposite signs, then
  - there is a point c on the real line for which $P(c)=0$.
  - that is, there is a root between a and b.
  - it is not necessary that there is only one root between such points; there may be $3,5,7,...$ roots; that is the number of real roots between a and b is odd and not even.

However, if some information about the roots are known, then we can try to find the other roots. For instance, if it is known that two of the roots of a polynomial equation of degree 6 with rational coefficients are $2+3i$ and $4-\sqrt{5}$, then we can immediately conclude that $2-3i$ and $4+\sqrt{5}$ are also roots of the polynomial equation. So dividing by the corresponding factors, we can reduce the problems into a problem of solving a second degree equation. In this section we learn some ways of finding roots of higher degree polynomials when we have some information.

3.7 Polynomials with Additional Information

Now we discuss a few additional information with which we can solve higher degree polynomials. Sometimes the additional information will directly be given, like, one root is $2+3i$. Sometimes the additional information like, sum of the coefficients is zero, have to be found by observation of the polynomial.

3.7.1 Imaginary or Surds Roots

If $\alpha+i\beta$ is an imaginary root of a quartic polynomial with real coefficients, then $\alpha-i\beta$ is also a root; thus $(x-(\alpha+i\beta))$ and $(x-(\alpha-i\beta))$ are factors of the polynomial; hence their product is a factor; in other words, $x^2-2\alpha x+\alpha^2+\beta^2$ is a factor; we can divide the polynomial with this factor and get the second degree quotient which can be solved by known techniques; using this we can find all the roots of the polynomial.

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